

CMSC 104 Section 2

Fall 2026

Classwork #5

While Loops

In-class date: October 7 2026

Due date: before midnight, October 14, 2026

Assignment:

You are going to get practice using “while” loops. Use the starter code below (also available as a file on both Blackboard and github), and solve all three parts.

```
# Classwork_5.py
# This file contains the "starter code" for Classwork 5
# You must complete all three parts of the assignment

"""
Part 1: User input.
The objective of this part is to use a "while" loop to control user input.
You will ask a user to input a test score, which is to be an integer. But
starting now, you will check for errors.
You will use the built-in Python method "isdigit()" to determine if the user
has actually input an integer.
As long as the user has not entered an integer, you must disregard the input
and ask the user to try again

"""

def get_user_input():
    score = ??? # ask the user to input a test score, which must be an integer
    while not(score.isdigit()): # score.isdigit() = True if score is an
integer; False otherwise
        print (???) # print out an error message telling the user that the
entry was NOT an integer
        score = ??? # ask the user to re-enter a score
    return ??? # you want to return the score
```

```

if __name__ == "__main__":
    # There are 35 students in the class. You have to call the function
    get_user_input() 35 times to get all the scores
    for i in range(0,35):
        s = get_user_input()
        print ("Score: ", i, " is ", s)

"""
Part 2: infinite loops
The following code contains an infinite loop. Your task is to rewrite
this code so it does NOT contain an infinite loop. The loop must run exactly
100 times.
Note: you MAY NOT change the while loop into a for loop.
"""

```

```

sum == 0
i == 0
while (i%2)**3 != 1:
    sum += i
    i += 4
print ("The answer is ", sum)

```

```

"""
Part 3: Sentinel loops

```

A sentinel while loop is a loop that runs until a certain value - called a sentinel - is encountered. In this case, the sentinel can be EITHER an uppercase "Q" OR a lowercase "q". Your task is to write a sentinel while loop that asks the user to input a student's name. The loop should run until the sentinel is encountered. That is, it should stop when the user types either a "Q" or a "q". Hint: this would be a really good place to use either the built-in string method .upper() or .lower()

```

"""

#first prompt the user for the first student's name.
# remember to tell the user that entering 'Q' or 'q' as the name will end the
program.
student_name = input(???)
while (???): # check to see if the user has entered the sentinal value. If
not, print the name.
    print(student_name)
    #now, prompt for the next student_name. Remind the user how to end the
program

```

```
print ("Thank you for using this program.")
```

Reminder

Assignments are your own effort. Do not share your code..

Grading Rubric

Part 1 works correctly: 40 points

Part 2 no longer contains an infinite loop and runs exactly 100 times: 20 points

Part 3 works correctly: 40 points

What to Submit

Your saved Python script.